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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - JAN'2025

COMMON TO B.E – CSE / B.TECH – IT / AIDS & M.TECH - CSE

20EE262 - BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING

Answer ALL questions

Duration: 2 Hrs 30 min

Maximum: 70 Marks

PART - A

(5x 1 = 5 Marks)

1. **CO1** Resistance of a wire is r ohms. The wire is stretched to double its length, then its resistance in ohms is _____.
a) $r/2$ b) $4r$ c) $2r$ d) $r/4$
2. **CO1** Kirchhoff's second law is based on law of conservation of _____.
a) charge b) energy c) momentum d) mass
3. **CO2** The transformers are rated in _____.
a) kVA b) kW c) kV d) kA
4. **CO3** In a 3-phase power measurement by two wattmeter method, both the watt meters had identical readings. The power factor of the load was _____.
a) unity b) 0.8 lagging c) 0.8 leading d) zero
5. **CO4** For an n-channel FET, What is the direction of current flow?
a) Source to drain b) Drain to source c) Gate to source d) Gate to drain

PART - B

(5x 3 = 15 Marks)

6. **CO1** How does an inductor act at $t=0^+$ and $t=\infty$?
7. **CO1** State Thevenin's theorem.
8. **CO2** State the principle of operation of transformer.
9. **CO3** What is the value of series resistance to be used to extent '0' to 200V range of $20,000\Omega/\text{volt}$ voltmeter to 0 to 2000 volt?
10. **CO4** Compare Avalanche breakdown with Zener breakdown.

PART - C

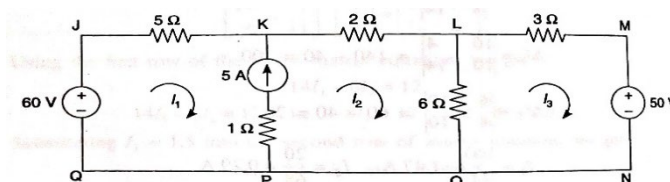
(5 x 10 = 50 Marks)

Q.No: 11 Compulsory Question

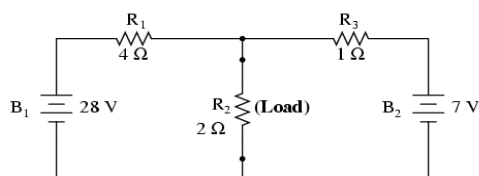
Q.No: 12 to 17 Answer any FOUR Questions

Compulsory Question:

11. CO1 Determine the current through 2Ω resistance in the network given below (10)
by mesh analysis method.



12. CO1 Obtain the Norton equivalent for the circuit given below and hence (10)
determine the current through the load resistance.



13. CO2 The coefficient of coupling between two coils is 0.6. There are 600 turns (10)
in coil 1. The total flux of coil 1 is 0.5mWb , when current in this coil is 4A. When the current is changed from 4A to 0A in 4milliseconds, the voltage induced in coil 2 is 50V. Calculate the self inductance and mutual inductance of the coils and the number of turns in coil 2.
14. CO3 Explain with necessary diagrams the construction and working of PMMC (10)
instruments.
15. CO3 Explain the three phase power measurement using two wattmeter method (10)
and derive the expression for power and power factor.
16. CO4 Explain the input and output characteristics of NPN transistor in CE (10)
configuration.
17. CO4 With necessary diagram explain the working of depletion type N channel (10)
MOSFET and draw its characteristics.
